Word Level Autoregressive Limerick Generation System

Foundations of Generative AI ICT4403 Assignment 2

Student name	Pranav Joshi
Registration number	230953248
Assignment title	Foundations of Generative AI Assignment 2
System title	Word Level Limerick Generation with RNN LSTM and GRU
Dataset	Limerick Dataset OEDILF v3 - https://doi.org/10.5281/zenodo.5722527
GitHub repository	https://github.com/mizar-0/RNN-LSTM-GRU-Comparison

Executive summary

Four word-level autoregressive models were trained on the filtered OEDILF limerick corpus under the same data split and core hyperparameters. GRU produced the lowest held-out perplexity at 45.30 and the highest top-1 and top-5 accuracy. In the generation sweep, GRU at temperature 0.3 produced five lines in every sample, had the strongest automatic AABBA rhyme score at 46.7%, and achieved a combined manual score of 3.0 out of 5. This configuration is the final recommendation, with the caveat that the manual sample was small and all generated models remained well below the human-reference rhyme score.

1 Problem and objective

The task was to build an autoregressive text-generation system using Simple RNN, Bidirectional RNN, LSTM, and GRU architectures. Each model predicts the next word from its preceding context and generates a poem one token at a time from a seed sequence. The comparison covers predictive performance, computational cost, generated-text quality, and the effect of sampling temperature.

2 Dataset and preprocessing

The Limerick Dataset OEDILF v3 was obtained from Zenodo under the CC BY 3.0 US licence. The source contains 98,454 records. After filtering by the supplied limerick label, 65,886 poems remained. Splitting was performed by poem before sequence construction, using 80% for training and 10% each for validation and test. This prevents token windows from the same poem from appearing in more than one partition.

Text was lowercased and tokenised into words and punctuation. Each poem received beginning-of-sequence, end-of-line, and end-of-sequence tokens. The training vocabulary was capped at 10,000 tokens and retained pad and unknown tokens. A 64-token left context was paired with each target token, yielding 2,229,796 training, 279,120 validation, and 278,760 test examples.

3 Model architectures

All four models shared the same input representation and output layer. A 128-dimensional embedding maps token indices to dense vectors. The recurrent layer uses 256 hidden units and dropout of 0.3. A linear projection then maps the recurrent representation to the 10,000-token vocabulary.

Model	Recurrent layer	Output width	Sequence representation	Parameters
Simple RNN	RNN 256	256	Final forward hidden state	3.95M
BiRNN	Bidirectional RNN 256	512	Concatenated forward and backward states	6.61M
LSTM	LSTM 256	256	Final hidden state	4.25M
GRU	GRU 256	256	Final hidden state	4.15M

4 Experimental protocol

The models used Adam with a learning rate of 0.001, batch size 512, context length 64, and random seed 42. Training was capped at 40 epochs. Validation-loss early stopping used a patience of five epochs, and the lowest-validation-loss checkpoint was retained. The final checkpoint for each model was evaluated once on the held-out test set. Because early stopping is part of the protocol, models trained for different numbers of epochs; the comparison uses each model's best validated checkpoint rather than its final epoch.

5 Training behaviour

The learning curves show that LSTM and GRU reached lower validation loss than the two vanilla recurrent models. Simple RNN stopped after 22 epochs with its best checkpoint at epoch 17. BiRNN stopped after 33 epochs with its best checkpoint at epoch 28. LSTM stopped after 26 epochs with its best checkpoint at epoch 21. GRU reached the 40-epoch cap and its best checkpoint was the last epoch, indicating that it had not triggered early stopping.

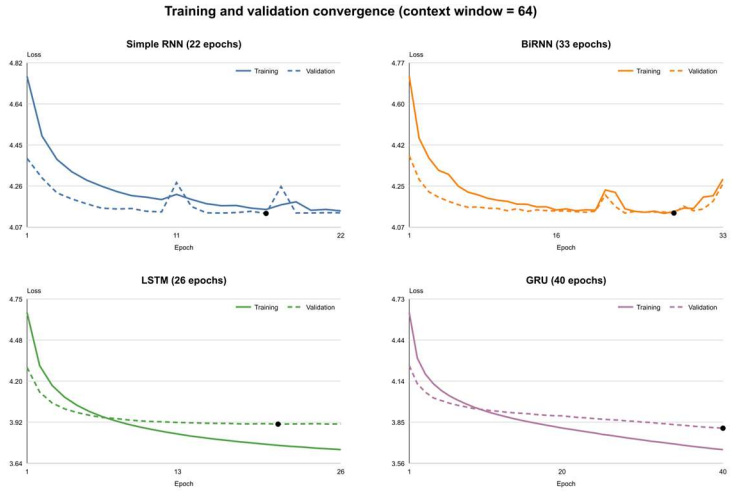


Figure 1 Training and validation loss for the final context length 64 runs

6 Held out test results

Model	Epochs	Parameters	Test PPL	Top 1	Top 5	Time
Simple RNN	22	3.95M	62.79	31.94%	50.56%	21.8 min
BiRNN	33	6.61M	62.81	32.00%	50.61%	51.7 min
LSTM	26	4.25M	50.03	34.14%	53.55%	29.9 min
GRU	40	4.15M	45.30	34.84%	54.79%	47.0 min

GRU achieved the strongest predictive result. Its perplexity of 45.30 was 9.5% lower than LSTM and about 27.9% lower than Simple RNN. It also obtained the highest top-1 accuracy at 34.84% and top-5 accuracy at 54.79%. LSTM was the best time-quality compromise: it improved substantially on both vanilla models while training 17.1 minutes faster than GRU. BiRNN increased parameter count and training time without improving perplexity over Simple RNN.

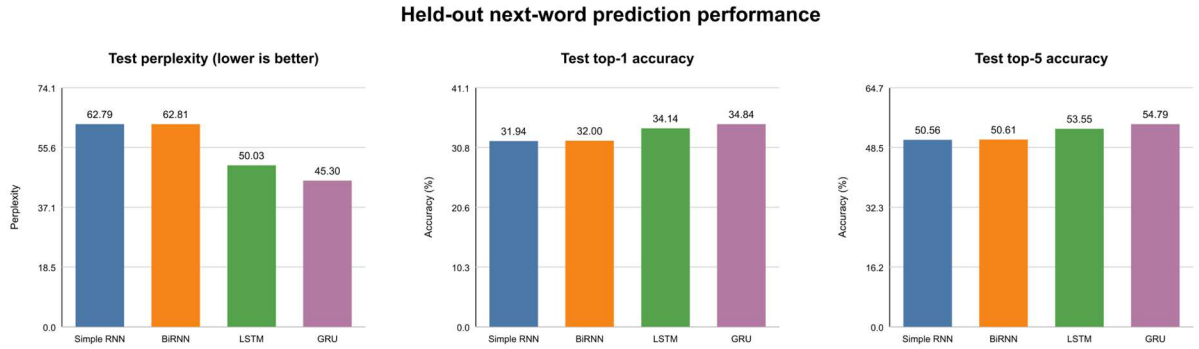


Figure 2 Held-out perplexity and next-word accuracy

7 Generation sweep

The generation sweep produced 7,200 poems from four models, six temperatures, and 300 poems per model-temperature cell. All models used the same five prompts. Automatic metrics covered termination, exact five-line form, pronunciation-dictionary coverage, AABBA rhyme, repeated trigrams, immediate repetitions, copied training lines, distinct n-grams, and unique-poem share. Metric uncertainty was estimated with 2,000 bootstrap resamples.

Temperature controlled a clear structure-diversity trade-off. Higher values increased lexical diversity but reduced five-line form and rhyme adherence. At temperature 0.3, GRU generated exactly five lines for 100% of samples and obtained the strongest AABBA rhyme score at 46.7%. The held-out human reference had an approximately 99.9% rhyme score, so the automatic result should be read as the best among the trained models rather than human-level poetic form.

8 Temperature effects

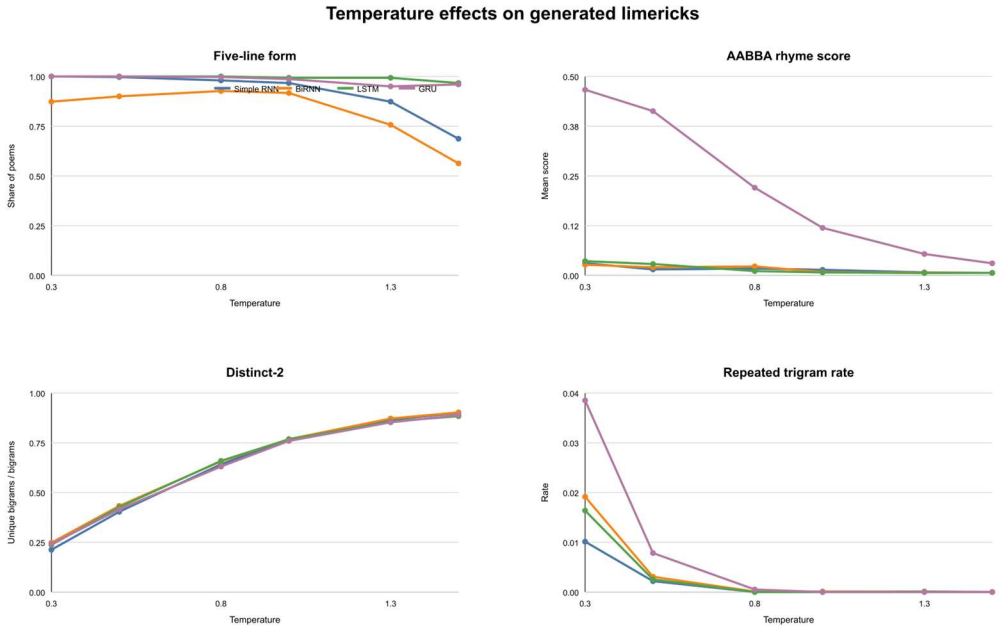


Figure 3 Form quality and diversity across sampling temperatures

Low-temperature decoding repeatedly selected high-probability words and therefore preserved structure more reliably. As temperature increased, the distribution flattened and the models selected more varied tokens. This improved distinct n-gram measures, but it also increased malformed line endings and weakened rhyme. Temperature 0.3 is appropriate for the assignment's structured limerick objective; temperatures from 0.8 to 1.0 are more suitable when variety matters more than consistent form.

9 Blinded manual evaluation

The blinded sheet contained 80 poems: four models, four temperatures, and five poems per cell. The rater scored coherence and contextual consistency from 1 to 5 without seeing model or temperature labels. GRU at temperature 0.3 achieved a combined score of 3.0 out of 5 with a 95% bootstrap interval of 2.6 to 3.4. LSTM at temperature 0.3 had the same combined point estimate, but GRU had a higher contextual-consistency score at 3.4 and stronger automatic rhyme evidence. Manual scores fell at high temperature for every model.

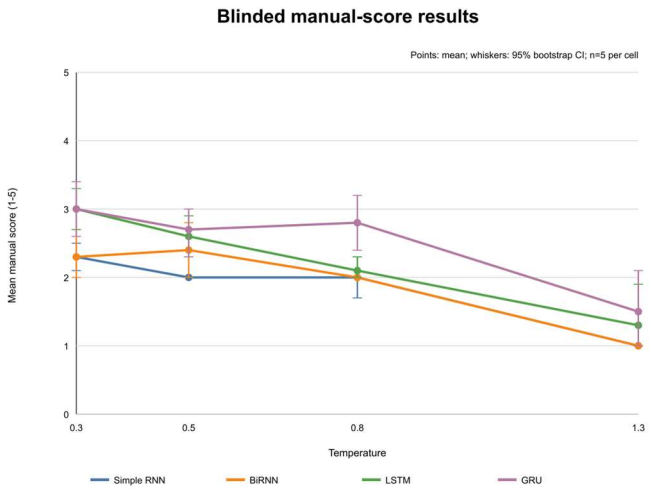


Figure 4 Mean blinded manual scores by model and temperature

10 Model selection

Model	Prediction	Training cost	Generated form	Decision
Simple RNN	Weak	Lowest	Moderate at low T	Useful baseline
BiRNN	Weak	Highest	Moderate at low T	Not selected
LSTM	Strong	Moderate	Strong at T 0.3	Best cost-quality alternative
GRU	Best	High	Best at T 0.3	Selected final model

The final system uses GRU with temperature 0.3. This choice is supported by the best held-out perplexity and accuracy, perfect five-line completion in the sweep, the strongest automatic rhyme score, and the highest contextual-consistency rating. LSTM at temperature 0.3 remains a credible alternative when shorter training time is more important than the last increment in predictive quality.

11 Limitations

The study used one random seed, so training variance is unknown. Early stopping produced different epoch counts, which is expected for validation-based model selection but prevents a strict equal-update comparison. GRU was still improving at the epoch cap and might benefit from a longer capped run. The rhyme metric depends on coverage in the CMU Pronouncing Dictionary. Manual evaluation used one rater and five poems per cell; overlapping intervals should not be interpreted as precise rankings. Perplexity measures next-token prediction and does not directly measure humour, narrative coherence, or originality.

12 Reproducibility

The repository contains preprocessing and training code, final model checkpoints, run metadata, learning curves, test metrics, the 7,200-sample generation sweep, automatic metric outputs, the blinded scoring workbook, and the unblinded manual-score summary. The final configuration uses seed 42 and records the data split, vocabulary size, context length, optimiser settings, epoch history, and checkpoint selection for each model.

13 Conclusion

Gating was the main architectural advantage in this experiment. LSTM and GRU both improved substantially on Simple RNN and BiRNN, while bidirectionality alone added compute without improving held-out perplexity. GRU at temperature 0.3 provides the strongest overall submission result. It should be reported with the manual-sample and rhyme-metric limitations above, and without claiming human-level poetic quality.

References

Limerick Dataset OEDILF v3. Zenodo. https://doi.org/10.5281/zenodo.5722527

CMU Pronouncing Dictionary. Carnegie Mellon University. http://www.speech.cs.cmu.edu/cgi-bin/cmudict

PyTorch. https://pytorch.org